

Anti-Claudin Treatments in Gastroesophageal Adenocarcinoma: Mainstream and Upcoming Strategies.

PMID: 37109309 | Indexed in PubMed

Claudins (CLDNs) are a multigene family of proteins and the principal components of tight junctions (TJs), which normally mediate cell-cell adhesion and selectively allow the paracellular flux of ions and small molecules between cells. Downregulation of claudin proteins increases the paracellular permeability of nutrients and growth stimuli to malignant cells, which aids the epithelial transition. Claudin 18.2 (CLDN18.2) was identified as a promising target for the treatment of advanced gastroesophageal adenocarcinoma (GEAC), with high levels found in almost 30% of metastatic cases. CLDN18.2 aberrations, enriched in the genomically stable subgroup of GEAC and the diffuse histological subtype, are ideal candidates for monoclonal antibodies and CAR-T cells. Zolbetuximab, a highly specific anti-CLDN18.2 monoclonal antibody, demonstrated efficacy in phase II studies and, more recently, in the phase III SPOTLIGHT trial, with improvements in both PFS and OS with respect to standard chemotherapy. Anti-CLDN18.2 chimeric antigen receptor (CAR)-T cells showed a safety profile with a prevalence of hematologic toxicity in early phase clinical trials. The aim of this review is to present new findings in the treatment of CLDN18.2-positive GEAC, with a particular focus on the monoclonal antibody zolbetuximab and on the use of engineered anti-CLDN18.2 CAR-T cells.

Extent and clinical significance of the therapy-relevant tight junction protein Claudin 18.2 in pancreatic ductal adenocarcinoma - real-world evidence.

PMID: 38917592 | Indexed in PubMed

Pancreatic ductal adenocarcinoma (PDAC) is associated with poor prognosis, wherefore targeted therapies have experienced increasing interest. Zolbetuximab is a novel targeted therapy under investigation in patients with PDAC and targets Claudin 18.2 (CLDN18.2), which is a component of tight junctions and is of significance in various solid tumors. As its role in PDAC is not definitively elucidated, this study aims to clarify the significance of CLDN18.2 expression in PDAC in a real-world setting. All patients (n = 309) were recruited at one of the PANCALYZE study centers and received pancreatic resection with curative intention. Paraffin samples were analyzed using an antibody against CLDN18.2, which is known to be comparable to the antibody used by the SPOTLIGHT and GLOW studies. 94 PDACs are positive for CLDN18.2 (30.4 %). Positive CLDN 18.2 expression was associated with significantly better cancer differentiation ($p < 0.001$). Patients with positive CLDN18.2 expression showed significantly better overall survival when compared to patients with negative expression (median OS: 30 versus 18 months, $p = 0.003$). Additionally, in multivariable analyses, CLDN18.2 expression was identified as an independent factor for better survival in patients with PDAC (HR = 0.686, 95 %CI = 0.492-0.956, $p = 0.026$). Significant improvement in survival could be demonstrated by adding Zolbetuximab to known chemotherapy regimes in patients with gastro-esophageal junction adenocarcinoma with at least 75 % CLDN18.2 positive cancer cells. Our findings demonstrate, that 30.4 % of the included patients with PDAC would potentially be eligible for therapy with Zolbetuximab in a real-world patient cohort. Results of trials targeting Claudin 18.2 are pending in patients with PDAC.

Comparison of Claudin 18.2 expression in primary tumors and lymph node metastases

in Japanese patients with gastric adenocarcinoma.

PMID: 31087075 | Indexed in PubMed

The monoclonal antibody zolbetuximab (formerly IMAB362), which is being developed as a potential treatment for gastric cancer (GC), targets Claudin 18.2 (CLDN18.2), a GC biomarker. This study aimed to determine the prevalence of CLDN18.2 in primary tumors and lymph node (LN) metastases of Japanese patients with GC. CLDN18.2 expression was investigated in tissue samples from patients with gastric adenocarcinoma archived at Kurume University Medical Center, Japan, between 2000 and 2012. Expression of CLDN18.2 in tumor samples was evaluated by immunohistochemistry using the same detection antibody (43-14A) and assay used in the FAST clinical trial (NCT01630083), a phase 2 randomized trial that compared the safety and antitumor activity of the zolbetuximab-chemotherapy combination with chemotherapy alone. Samples showing any specific staining with $\geq 1+$ intensity were defined as CLDN18.2-positive. Of 263 samples analyzed (134 primary gastric tumors and corresponding LN metastases; 128 primary tumors only; one LN metastases only), CLDN18.2 was detected in 87% ($n = 228/262$) of all primary tumors and 80% ($n = 108/135$) of LN metastases. Moderate-to-strong CLDN18.2 expression ($\geq 2+$ membrane staining intensity in $\geq 40\%$ of tumor cells [FAST eligibility criterion]) was observed in 52% ($n = 135/262$) of primary tumors and 45% ($n = 61/135$) of (LN) metastases. CLDN18.2 expression was significantly higher in GCs of the diffuse histological subtype per Lauren classification and in high grade (G3) tumors. The high prevalence of CLDN18.2 among Japanese patients with GC supports the therapeutic assessment of zolbetuximab in this population.

ILUSTRO: Phase II Multicohort Trial of Zolbetuximab in Patients with Advanced or Metastatic Claudin 18.2-Positive Gastric or Gastroesophageal Junction Adenocarcinoma.

PMID: 37490286 | Indexed in PubMed

Zolbetuximab, an IgG1 monoclonal antibody, binds to claudin 18.2 (CLDN18.2) and mediates tumor cell death through antibody-dependent cellular cytotoxicity and complement-dependent cytotoxicity. We sought to examine zolbetuximab combinations in CLDN18.2-positive HER2-negative gastric/gastroesophageal junction (G/GEJ) adenocarcinoma. This phase II study assessed efficacy and safety of zolbetuximab, alone or with modified FOLFOX6 (mFOLFOX6) or pembrolizumab, in CLDN18.2-positive advanced/metastatic G/GEJ adenocarcinoma. Patients received zolbetuximab as monotherapy in third/late-line (Cohort 1A, $n = 30$), with mFOLFOX6 in first-line (Cohort 2, $n = 21$), or with pembrolizumab in third/late-line (Cohort 3A, $n = 3$) treatment. The primary endpoint for Cohort 1A was objective response rate (ORR). Key secondary endpoints were ORR (Cohorts 2 and 3A), overall survival (OS; Cohort 1A), and progression-free survival (PFS) and safety (all cohorts). ORR was 0% in Cohorts 1A and 3A, and 71.4% [95% confidence interval (CI), 47.82-88.72] in Cohort 2. Median PFS was 1.54 months (95% CI, 1.31-2.56) in Cohort 1A, 2.96 months (95% CI, 1.48-4.44) in Cohort 3A, and 17.8 months (95% CI, 8.05-25.69) in Cohort 2. Median OS in Cohort 1A was 5.62 months (95% CI, 2.27-11.53). Gastrointestinal adverse events occurred across cohorts [nausea, 63%-90% (grade ≥ 3 , 4.8%-6.7%) and vomiting, 33%-67% (grade ≥ 3 , 6.7%-9.5%)]. Zolbetuximab plus mFOLFOX6 demonstrated promising efficacy in previously untreated patients with CLDN18.2-positive G/GEJ adenocarcinoma. These data support the first-line development of zolbetuximab in patients whose tumors are CLDN18.2-positive. Across cohorts, zolbetuximab treatment was tolerable with no new safety signals.

Preclinical characterization of an mRNA-encoded anti-Claudin 18.2 antibody.

PMID: 37860278 | Indexed in PubMed

IMAB362/Zolbetuximab, a first-in-class IgG1 antibody directed against the cancer-associated gastric-lineage marker CLDN18.2, has recently been reported to have met its primary endpoint in two phase 3 trials as a first-line treatment in combination with standard of care chemotherapy in CLDN18.2-positive Her2 negative advanced gastric cancer. Here we characterize the preclinical pharmacology of BNT141, a nucleoside-modified RNA therapeutic encoding the sequence of IMAB362/Zolbetuximab, formulated in lipid nanoparticles (LNP) for liver uptake. We show that the mRNA-encoded antibody displays a stable pharmacokinetic profile in preclinical animal models, mediates CLDN18.2-restricted cytotoxicity comparable to IMAB362 recombinant protein and inhibits human tumor xenograft growth in immunocompromised mice. BNT141 administration did not perpetrate mortality, clinical signs of toxicity, or gastric pathology in animal studies. A phase 1/2 clinical trial with BNT141 mRNA-LNP has been initiated in advanced CLDN18.2-expressing solid cancers (NCT04683939).

Effect of antiemetics on zolbetuximab-induced gastric injury and emesis in ferrets.

PMID: 39313274 | Indexed in PubMed

Claudin-18 splice variant 2 (CLDN18.2), a tight junction protein, is a highly cell type-specific antigen that is expressed by differentiated gastric mucosa cells. The expression of CLDN18.2 in gastric mucosa cells may be retained upon malignant transformation and is displayed on the surface of several tumors, including gastric/gastroesophageal junction adenocarcinoma. Zolbetuximab is a genetically engineered, highly purified chimeric (mouse/human IgG1) antibody directed against CLDN18.2. Nausea and vomiting were observed as adverse events of zolbetuximab. To investigate the mechanism of nausea and vomiting in humans, we evaluated emesis (retching and vomiting) and conducted histopathologic assessment in ferrets after the administration of zolbetuximab. Emesis was frequently observed in all ferrets treated with zolbetuximab in the first hour after administration. Histopathologic assessment revealed the surface of the gastric mucosa was the primary site of emesis-associated tissue damage. The effect of antiemetics (dexamethasone, ondansetron, fosaprepitant, and olanzapine) on emesis induced by zolbetuximab was investigated. Fosaprepitant showed suppressive effects on emesis, and use of dexamethasone or concomitant use of fosaprepitant with other antiemetics tended to alleviate gastric tissue damage. The onset of emesis in humans receiving zolbetuximab may be associated with damage in the gastric mucosa, and antiemetics may mitigate gastrointestinal adverse events.

Molecular SPECT/CT Profiling of Claudin18.2 Expression In Vivo: Implication for Patients with Gastric Cancer.

PMID: 38843446 | Indexed in PubMed

Zolbetuximab (IMAB362), a monoclonal antibody targeting Claudin18.2 (CLDN 18.2), demonstrates a significant clinical benefit in patients with advanced gastroesophageal cancers. The noninvasive assessment of CLDN18.2 expression through molecular imaging offers a potential avenue for expedited monitoring and the stratification of patients into risk groups. This study elucidates that CLDN18.2 is expressed at a noteworthy frequency in primary gastric cancers and their metastases. The iodogen

method was employed to label IMAB362 with $^{123}\text{I}/^{131}\text{I}$. The results demonstrated the efficient and reproducible synthesis of ^{123}I -IMAB362, with a specific binding affinity to CLDN18.2. Immuno-single-photon emission computed tomography (SPECT) imaging revealed the rapid accumulation of ^{123}I -IMAB362 in gastric cancer xenografts at 12 h, remaining stable for 3 days in patient-derived tumor xenograft models. Additionally, tracer uptake of ^{123}I -IMAB362 in MKN45 cells surpassed that in MKN28 cells at each time point, with tumor uptake correlating significantly with CLDN18.2 expression levels. Positron emission tomography/computed tomography imaging indicated that tumor uptake of ^{18}F -FDG and the functional/viable tumor volume in the ^{131}I -IMAB362 group were significantly lower than those in the ^{123}I -IMAB362 group on day 7. In conclusion, ^{123}I -IMAB362 immuno-SPECT imaging offers an effective method for direct, noninvasive, and whole-body quantitative assessment of tumor CLDN18.2 expression in vivo. This approach holds promise for accelerating the monitoring and stratification of patients with gastric cancer.

Efficacy and safety of zolbetuximab for first-line treatment of advanced Claudin 18.2-positive gastric or gastro-esophageal junction adenocarcinoma: a systematic review and meta-analysis of randomized controlled trials.

PMID: 37886169 | Indexed in PubMed

Zolbetuximab is a "first-in-class" chimeric IgG1 monoclonal antibody targeting Claudin18.2 (CLDN 18.2). In recent years, several important trials have been published showing that zolbetuximab is associated with improved prognosis in patients with advanced gastric or gastro-esophageal junction (G/GEJ) adenocarcinoma. This promises great change to the current treatment landscape. Therefore, we conducted a systematic review and meta-analysis to evaluate the efficacy and safety of zolbetuximab for first-line treatment of advanced CLDN 18.2-positive G/GEJ adenocarcinoma. The following databases were searched for relevant studies: PubMed, EMBASE, and Cochrane library (updated 10 June 2023). All randomized trials comparing zolbetuximab plus chemotherapy versus first-line chemotherapy alone for first-line treatment of advanced CLDN 18.2-positive G/GEJ adenocarcinoma were eligible for inclusion. Data were analyzed using Review Manager 5.4.1 (Cochrane collaboration software). Primary outcomes and measures included overall survival (OS), progression-free survival (PFS), objective response rate (ORR), and adverse events (AEs). This systematic review and meta-analysis included three randomized controlled studies involving 1,402 patients (699 receiving zolbetuximab plus chemotherapy and 703 receiving chemotherapy alone). Compared with chemotherapy alone, zolbetuximab plus chemotherapy significantly improved OS (HR = 0.73; 95% CI: 0.68-0.84) and PFS (HR = 0.64; 95% CI: 0.50-0.82), but did not result in a higher ORR (RR = 0.92; 95% CI: 0.82-1.03). Further analysis of CLDN 18.2 expression showed a more significant benefit for OS (HR = 0.69; 95% CI: 0.55-0.87; $p = 0.002$) and PFS (HR = 0.61; 95% CI: 0.44-0.84; $p = 0.003$) from zolbetuximab in patients with high expression, while there was significant benefit in patients with lower expression. In terms of AEs, zolbetuximab plus chemotherapy was associated with higher risk of grade 3 and higher AEs, but increased risk of nausea and vomiting were more common. This systematic review and meta-analysis revealed that the effect of zolbetuximab plus chemotherapy was superior to that of chemotherapy alone for first-line treatment of advanced CLDN 18.2-positive G/GEJ adenocarcinoma. Thus, zolbetuximab plus chemotherapy represents a new first-line treatment for these patients. Zolbetuximab plus chemotherapy was associated with higher risk of grade 3 and higher AEs, but was generally manageable. <https://www.crd.york.ac.uk/prospero>, identifier (CRD42023437126).

Evaluation and reflection on claudin 18.2 targeting therapy in advanced gastric cancer.

PMID: 32410803 | Indexed in PubMed

Claudin 18.2 (CLDN18.2) is a tight-junction protein. CLDN18.2-targeting strategy has cut a striking figure in CLDN18.2 positive patients with advanced gastric cancer. Zolbetuximab, the CLDN18.2 antibody, obtained a better clinical benefit in patients compared with the controlled. In phase II trials, combination treatment of epirubicin, oxaliplatin and capecitabine (EOX) + zolbetuximab achieved the optimal effects of overall survival which extended to 13.2 months with tolerable safety events, indicating its greater potential playing the second promising target in gastric cancer. This review will reveal the definitive clinical benefit CLDN18.2-targeting therapies have achieved and update the highlighting development (like chimeric antigen receptor T-cell immunotherapy) to CLDN18.2 positive patients. We then focus on 10 questions arisen from recent progress and anticipate to provide a future perspective for novel cancer treatment.

Claudins in vulvar cancer - from epithelial barrier to potential tumor-agnostic cancer therapy.

PMID: 39722127 | Indexed in PubMed

The immunohistochemical expression of various members of the claudin family has already been studied in pathological affections of the vulva whether to differentiate precancerous lesions from vulvar squamous cell carcinoma or in inflammatory conditions such as lichen sclerosus. From an oncological perspective, however, immunohistochemical analysis of claudin 18.2 protein expression has become increasingly clinically relevant nowadays since the impressive therapeutic benefits of the claudin 18.2 antibody zolbetuximab have been widely recognized. Systematic studies evaluating its expression, including in vulvar cancer, are needed to understand whether such treatment strategies may eventually benefit patients with vulvar neoplasia.